

Question 1: Object-Oriented Programming (OOP)

[10 marks]

Introduction: (2 marks)

OOP stands for Object Oriented Programming. It is a way of writing programs using objects. An object has data and can do things (methods). Popular languages like Java and Python use OOP.

Core Concepts: (4 marks)

OOP has four pillars. Encapsulation keeps data private inside a class and gives public methods to access it. Inheritance lets a child class get properties from a parent class so we can reuse code. Polymorphism means the same method name can work differently in different classes. Abstraction hides details and shows only what is needed through interfaces or abstract classes.

Examples: (2 marks)

A Dog class can inherit from Animal class. Dog has its own speak() method that overrides Animal's speak(). A BankAccount class has private balance and public deposit() and withdraw() methods to encapsulate the data.

Conclusion: (2 marks)

OOP helps us write programs that are easier to manage. Code reuse through inheritance saves time. The four pillars make software modular and maintainable.

Question 2: Database Normalization

[10 marks]

Introduction: (2 marks)

Database normalization organizes tables to reduce redundancy and improve integrity. It splits large tables into smaller related tables. This prevents problems during insert update and delete operations.

Core Concepts: (4 marks)

1NF means each cell has one value and no repeating groups. 2NF means we are in 1NF and every non-key column depends on the whole primary key. 3NF means no non-key column depends on another non-key column (no transitive dependency). BCNF is stronger than 3NF where every determinant is a candidate key.

Examples: (2 marks)

StudentCourse table with StudentID, StudentName, CourseID, CourseName. StudentName only depends on StudentID so this is a partial dependency violating 2NF. We split into Student(StudentID, StudentName) and Course(CourseID, CourseName).

Conclusion: (2 marks)

Normalization makes databases clean and consistent. Higher normal forms reduce anomalies but may need more joins. It is essential for reliable database design.

Question 3: Operating System Process Management

[10 marks]

Introduction: (2 marks)

Process management is done by the operating system. A process is a running program. The OS handles many processes together by sharing CPU memory and I/O resources between them.

Core Concepts: (4 marks)

Process states are New Ready Running Waiting and Terminated. The PCB (Process Control Block) stores the process state PID and registers. Scheduling algorithms choose which process runs. FCFS runs processes in arrival order. Round Robin gives each process a fixed time quantum. SJF picks shortest job first.

Examples: (2 marks)

Round Robin with quantum 4ms: P1 24ms P2 3ms P3 3ms. Order is P1(4) P2(3) P3(3) P1(4) P1(4) P1(4) P1(4) P1(4). Context switch saves and loads PCB data.

Conclusion: (2 marks)

Process management keeps the CPU busy and fair. Good scheduling improves response time and throughput. The OS must prevent deadlocks and starvation.

Question 4: Computer Networks and the OSI Model

[10 marks]

Introduction: (2 marks)

A computer network connects devices so they can share data. The OSI model has 7 layers and is a standard for how data travels across a network. It was made by ISO in 1984.

Core Concepts: (4 marks)

Layer 7 Application: HTTP FTP DNS for user apps. Layer 6 Presentation: encrypts and translates data. Layer 5 Session: manages sessions. Layer 4 Transport: TCP (reliable) and UDP (fast). Layer 3 Network: IP addresses and routing. Layer 2 Data Link: MAC addresses and error detection. Layer 1 Physical: cables and bits.

Examples: (2 marks)

When opening a website: DNS resolves the name. Browser sends HTTP request. TCP segments it. IP routes it through routers. Ethernet carries it over LAN. TCP three-way handshake: SYN SYN-ACK ACK before data is sent.

Conclusion: (2 marks)

The OSI model is a reference model for networks. TCP/IP is the real 4-layer model used on the internet. Understanding layers helps in troubleshooting network problems and designing protocols.